

ARCHIMEDES Preparing PNG Images for Submission

A guide for projects sharing image (PNG) data with ARCHIMEDES

ARCHIMEDES accepts images saved as PNG files, such as echocardiogram stills and ECG traces. Before your data can be ingested we need to know which participant and which visit each image belongs to, and what kind of image it is. The folder and file names carry all three.

You only need to provide three things

A folder tree organised by participant and visit, image files named for what they show, and two small tab-separated files: participants.tsv and a sessions.tsv per participant. You do not create any records in ARCHIMEDES; participant creation, identifiers and archiving to the platform are handled on our side.

1. Folder structure

Share your images with the ARCHIMEDES team in a folder named for your project. Inside it, use one folder per participant, and inside that one folder per visit. Place the image files for that visit inside the visit folder.

```
<project_name>/deidentified-raw/imaging/png/
├── participants.tsv
├── sub-ARCHI0001/
│   ├── sub-ARCHI0001_sessions.tsv
│   ├── ses-V01/
│   │   ├── echo_001.png
│   │   ├── echo_002.png
│   │   └── ecg_001.png
│   ├── ses-V02/
│   │   └── echo_001.png
│   └── sub-ARCHI0002/
│       ├── sub-ARCHI0002_sessions.tsv
│       ├── ses-V01/
│       │   └── echo_001.png
```

Both levels are fixed: the sub- and ses- prefixes tell us which part of a folder name is your identifier. Unlike DICOM there is no folder below the visit; the images sit directly inside it, and the file name says what each one is.

2. Naming rules

Identifiers are yours to choose. Keep them to letters and numbers; underscores and dashes cause problems downstream. The sub- and ses- prefixes are separate and always required.

Rule	Good	Avoid
No underscores in identifiers	sub-ARCHI0001	sub-ARCHI_0001, sub-ARCHI-0001
No spaces, slashes or quotation marks	sub-P014	sub-Patient 14
No names or identifying information	sub-UOHI0007	sub-Smith J
Same identifier for a participant across all their visits	sub-P014/ses-V01 and /ses-V02	sub-P014 then sub-P14
Visit labels short and consistent	ses-V01, ses-Baseline	ses-visit one

Image file names

Each file name begins with the type of image it is, followed by a number when a visit holds more than one of that type. We agree on the list of image type names with you during onboarding, so everyone sending data to your project uses the same words.

echo_001.png	first echocardiogram still
echo_002.png	second echocardiogram still
ecg_001.png	ECG trace

Use only names from that list. A file whose name we do not recognise is set aside and reported back to you rather than loaded.

Rule	Good	Avoid
Image type first, letters and numbers only	echo_001.png	Echo Study 1.png
Use the agreed name, not a description	ecg_001.png	twelve lead tracing.png
Pad numbers so they sort correctly	echo_001, echo_002, echo_010	echo_1, echo_2, echo_10

Visits are not created automatically

Visit labels must already be configured in ARCHIMEDES, and a folder using an unconfigured label is skipped. We set these up with you during onboarding; tell us the labels that fit how your study runs, and any new timepoints later.

3. The two spreadsheets

Build these in Excel or Google Sheets if that is easiest. What matters is how you save them: choose Tab Delimited Text (Excel) or Tab-separated values (Sheets), not comma-separated and not .xlsx.

participants.tsv — one row per participant

Location: the top level of your folder, alongside the sub- folders.

Column	Required	Format	Example
participant_id	Yes	Must match the sub- folder name	sub-ARCHI0001
dob	Yes	YYYY-MM-DD, YYYY-MM or YYYY	1948-03-22
sex	Yes	Male, Female or Other	Male
project	Preferred	Your study name in ARCHIMEDES	CardiacImaging
site	Preferred	Where the participant was seen	UOHI
cohort	Preferred	Group within the study	Control

participant_id	dob	sex	project	site	cohort
sub-ARCHI0001	1948-03-22	Male	CardiacImaging	UOHI	Control
sub-ARCHI0002	1962-11-08	Female	CardiacImaging	UOHI	Treatment

Only the year and month of the date of birth are stored, so send whatever precision you are comfortable with: a full date, YYYY-MM, or just YYYY. If you are not sure what to put for project, site or cohort, leave them out and ask the ARCHIMEDES technical support team.

sessions.tsv — one per participant

Location: inside that participant’s sub- folder, named after it.

For example sub-ARCHI0001_sessions.tsv. One row per visit. Only session_id is required; add any further columns useful to you.

Column	Required	Format	Example
session_id	Yes	Must match the ses- folder name	ses-V01
acq_date	Optional	YYYY-MM-DD, date of the visit	2024-11-19

4. About the images

What we read

A PNG carries no hidden information, so everything we need comes from the names you give us.

Where we read it	What it tells us
The sub- folder name	Your identifier for the participant, stored as the External ID. We create the participant record from it, and you can always find them by it
The ses- folder name	Which visit the image belongs to
The start of the file name	What kind of image it is, matched against the list agreed during onboarding

What we change

On our copy only; your originals are never modified. We rename our copy to the ARCHIMEDES convention and record the file name you gave it, so an image can always be traced back to your own export. The image itself is never altered.

What should not be visible in the image

This is the one part of a PNG submission with no equivalent in DICOM. A DICOM header can be de-identified with software; a PNG has no header at all. Everything it carries is what you can see, so anything identifying is printed into the picture itself and no tool can remove it automatically. Exports from ultrasound and echocardiography machines routinely print the following across the top of the frame:

- Patient name, initials or medical record number
- Date of birth, or the date and time of the examination
- Operator name, machine name or hospital name

5. What we do with the images

- Organise by participant and visit, and name each file for the type of image it holds.
- Every image in a visit folder is loaded, however many there are.
- A small preview is generated for each image so that it can be found and viewed in ARCHIMEDES.
- If your export does not fit this shape, talk to us before sending. There is usually a way to accommodate it, and we would rather adapt on our side than have you rename hundreds of files.